

8/24/26 Notes from Dan – Lensy

SEE 1 NOTE PER PAGE, BELOW

Note:

Two key themes require extra attention:

1. If user asks a question which does not 'fit' the prescribed AI Guide response format, there can be uncomfortable or rambling responses. Not every search will demand references to multiple standards. Some users just want a single specific answer which exists in only one standard; others will ask a question that can't be answered by our standards. We need an option to adapt
2. Title names, section names, and hierarchy is not always accurate. Is this useful reference for additional learning? [260823_Document headings and structure.docx](#)

DO 070. We've had a few users not understand that "From the Standard" is a drop-down. Can you try a different format? One example below, but I'm open to alternatives.

Sample Search:	What are considerations for lighting parking garages?
Goal:	Intuitive interface
Example:	See below

One thought: begin with the carrot facing *down* instead of facing right (i.e. lose the animation)

17.5.2.1

RP-8-25+E1

ANSI/IES RP-8-25+E1 Recommended Practice: Lighting Roadway and Parking Facilities, p. 389
IES Roadway Lighting Committee

> FROM THE STANDARD

Change to default:

17.5.2.1

RP-8-25+E1

ANSI/IES RP-8-25+E1 Recommended Practice: Lighting Roadway and Parking Facilities, p. 389
IES Roadway Lighting Committee

✓ FROM THE STANDARD

DO 071.

[See also DO 073] Section titles and section numbers are not always accurate.

Sample Search:

What is the ceiling uniformity ratio for indirect lighting in an office space

Goal:

Accuracy, legibility

Example:

See below

131 - PATTERNS OF LIGHT WITHIN THE OCCUPANTS' FIELD OF VIEW—SUCH AS WALLS AND

LP-1-24 Match: 73% Open in Library Save Search

ANSI/IES LP-1-24 Lighting Practice: Designing Quality Lighting for People and Buildings, p. 77

FROM THE STANDARD — 2 passages

131 Patterns of Light within the occupants' field of view—such as walls and

"[Section 131] ibility across the task plane. That is why library stack lighting, desktop lighting, industrial lighting, and parking lot lighting, for example, are often recommended with a uniformity ratio. Depending on the application, the rec- Figure 13-19 Overhead downlights produce poor uniformity on this workstation desktop 62 ommended uniformity ratio may be less than a certain maximum-to-min..."

p. 77 Open in Library

131 Patterns of Light within the occupants' field of view—such as walls and

"[Section 131] surrounding illuminance levels on the desk. In an outdoor parking lot, dark areas can obscure curbs and obstacles; IES recommends no more than a 15:1 maximum-to-minimum ratio on the parking surface.28 • After determining the basic lighting layout, the designer should do a mockup or perform a point- by-point illuminance calculation to identify problem areas. If there are any places wh..."

p. 77 Open in Library

- Should be

- 13.4 Light Distribution on Task Plane (Uniformity)

13.4 Light Distribution on Task Plane (Uniformity)

- Uniformity, the even distribution of illumina

A.1.1.3 - REGULAR AREA WITH SINGLE ROW OF INDIVIDUAL A.1.1.6 REGULAR AREA WITH UNIFORM INDIRECT LIGHTING.THE

LP-9-25 Match: 70% Open in Library Save Search

ANSI/IES LP-9-25 Lighting Practice: Upgrading Lighting Systems in Nonresidential Spaces, p. 68

IES Lighting Upgrades Committee

FROM THE STANDARD

A.1.1.3 Interior Measurements Average Illuminance on a Horizontal Plane.The Regular Area With Single Row of Individual A.1.1.6 Regular Area With Uniform Indirect Lighting.The

"A.1.1.3 Regular Area With Single Row of Individual A.1.1.6 Regular Area With Uniform Indirect Lighting.The Luminaires. The average illuminance, E_{avg} , in such a average illuminance, E_{avg} , in such a space (See Figure space (See Figure A-1c) can be determined from: A-1f) can be determined from: $Q(\frac{N-1}{N}) + P E_{avg} = ,$

$\frac{R(L-8)(W-8)+8 Q(L-8)+8 T(W-8)+64P}{N}$

p. 68 Open in Library

- Should be

- A.1.1.3 Regular Area With Single Row of Individual Luminaires

A.1.1.3 Regular Area With Single Row of Individual Luminaires. The average illuminance, E_{avg} , in such a space (See Figure A-1c) can be determined from:

$$E_{avg} = \frac{Q(N-1) + P}{N} ,$$

where:

N = number of luminaires

-

DO 072.

Formulae do not display properly. If we can display them accurately (perhaps as a screenshot?) great. If not, it would be better to return 'Open in Library to see formula' than to display them inaccurately.

Sample Search:

What is the ceiling uniformity ratio for indirect lighting in an office space

Goal:

Accuracy

Example:

See below

A.1.1.3 REGULAR AREA WITH SINGLE ROW OF INDIVIDUAL A.1.1.6 REGULAR AREA WITH UNIFORM INDIRECT LIGHTING.THE Document

LP-9-25 Match: 70% Open in Library + Save Search

ANSI/IES LP-9-25 Lighting Practice: Upgrading Lighting Systems in Nonresidential Spaces, p. 68
IES Lighting Upgrades Committee

FROM THE STANDARD

A.1.1.3 Interior Measurements > Average Illuminance on a Horizontal Plane.The Regular Area With Single Row of Individual A.1.1.6 Regular Area With Uniform Indirect Lighting.The

"A.1.1.3 Regular Area With Single Row of Individual A.1.1.6 Regular Area With Uniform Indirect Lighting.The Luminaires. The average illuminance, E_{avg} , in such a average illuminance, E_{avg} , in such a space (See Figure space (SeeFigure A-1c) can be determined from: A-1f) can be determined from: $E_{avg} = \frac{Q(N-1)+P}{N}$

$E_{avg} = \frac{Q(N-1)+P}{N}$

p. 68 Open in Library

A.1.1.3 Regular Area With Single Row of Individual Luminaires. The average illuminance, E_{avg} , in such a space (See **Figure A-1c**) can be determined from:

$$E_{avg} = \frac{Q(N-1)+P}{N},$$

where:

N = number of luminaires

Q = Average of measurements at stations q-1 through q-8 in four typical half bays located two on each side of the area

P = Average of measurements at stations p-1 and p-2 for two typical corner quarter bays

(Note: this formula has the added complexity of a column break)

An acceptable return would be a screenshot *or* a note "Open in Library to see formula"

[DO 072a]

Do not display formulae in the AI Guide. The text-only representation can be misleading and/or inaccurate.

Example search:

Q: What document supplies the calculation for LM-84 test data?

Search

 **AI Guide**

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

Beta

The calculation for LM-84 test data is a crucial aspect of projecting long-term luminous flux maintenance of LED lamps and luminaires. According to ANSI/IES TM-28-20, the calculation involves an exponential decay formula, as seen in Annex B of the standard. This formula, $\Phi(t)_{LM-84} = B_{LM-84} \cdot \exp(-t \alpha_{LM-84})$, is used to approximate the luminous flux maintenance over time. The standard provides guidance on how to derive this formula and apply it to LM-84 test data.

Instead, could just print “This formula is used to approximate...”

DO 073.

‘Document’ result cards (only): Let’s restructure results to 1 card per chapter (e.g. 4.3.3.1 and 4.2.4 both equal chapter 4). Then refine sort order to place best-quality results at top.

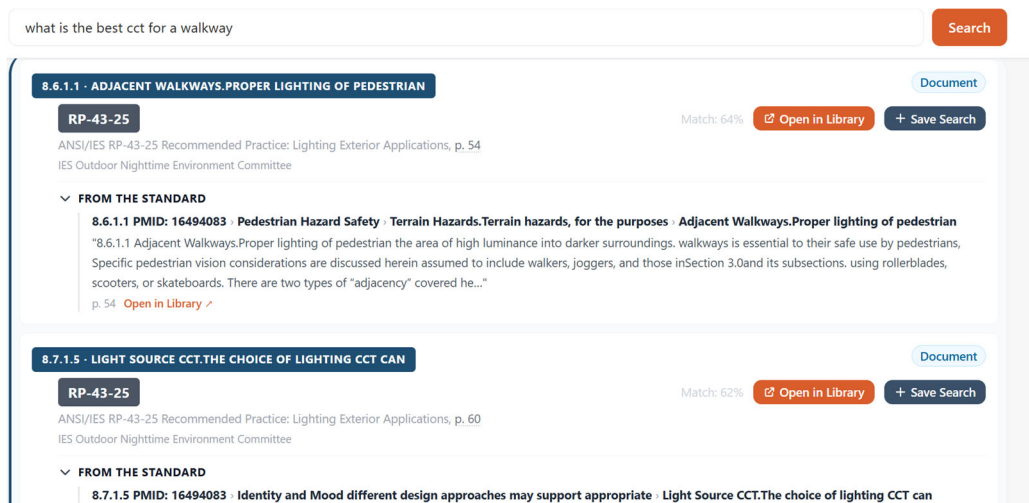
Sample Search:

What is the ceiling uniformity ratio for indirect lighting in an office space

Goal:

Shift highest-quality results to top of results, while retaining visual relationship within a document/chapter.

- Sort order for chapter cards: highest quality chapter results first, even if that differs from printed order. Allow ‘AI Guide’ to help curate priority order.
- Print *chapter name* in the blue band at the top of the card.
 - o Within each chapter, move any results to drop-down “From the Standard”
 - o “From the Standard” defaults to *expanded* instead of collapsed
 - o In ‘From the Standard’ the bolded text remains the section title as currently printed



Changes to:

Ch. 8 – Outdoor Lighting Design Process

RP-43-25

ANSI/IES RP-43-25 Recommended Practice: Lighting Exterior Applications, p. 54
IES Outdoor Nighttime Environment Committee

FROM THE STANDARD

8.7.1.5 Identity and Mood > Identity > Light Source CCT

8.6.1.1 Pedestrian Hazard Safety > Terrain Hazards > Adjacent Walkways

DO 074. ~~[Previously described in Teams chat]~~ Based on beta testing and market expectations, we need to do anything we can to truly harness AI assistance for locating and prioritizing result cards.

DO 062. [Re-posting with clarification] Document comparison tool still needs greater depth and detail:

- Permit much longer 'AI Document Comparison' response lengths (for these comparison responses only). Link to relevant pages in the current standard whenever possible.
- ~~[done]-Hyperlink to the deprecated standards in the responses (same as the current standards) for these responses~~
- When a section is mentioned (e.g. §8.7.2.4) hyperlink to the page in the document
- When available, present the section title adjacent to section number, and make all section numbers & titles **bold** font.
- Sort results in each category (new, updates, deletions) by section (or page number). **Let's make this a bulleted list under each category, structured by chapter**

Sample Search: What is new in the current version of ANSI/IES RP-43?
Goal: Greater depth; titles/sections easier to pick out for quick evaluation
Example: See below

If you need a verified comparison, please perform a manual review of both documents. **ANSI/IES RP-43-22** is deprecated and has been replaced by the current **ANSI/IES RP-43-25 Recommended Practice: Lighting Exterior Applications**. This comparison is between those two editions only.

What appears to be new

- The current edition, **ANSI/IES RP-43-25**, appears to cover outdoor lighting requirements in **§6.2** which is not found in the excerpts from the prior edition, **ANSI/IES RP-43-22**.
- The **§8.7.2.4** section in the current edition discusses considerations for lighting design, which is not present in the excerpts from the prior edition.
- The current edition also mentions the Five Principles for Responsible outdoor lighting in **§6.2**, which is not found in the prior edition.

- Green highlights: hyperlink & **bold**
- Orange highlights: hyperlink & **bold** & add section name when available.
 - o Format so each reference to a section begins with the section number & name for easy scanning, like a table of contents:

What appears to be new

- The current edition, **ANSI/IES RP-43-25**, appears to cover outdoor lighting requirements in **6.2 Outdoor Lighting Requirements**, which is not found in the excerpts from the prior edition, **ANSI/IES RP-43-22**.
- **Chapter 6.0 Community Planning**
 - **6.2 Outdoor Lighting Requirements:** Mentions the Five Principles for Responsible Outdoor Lighting; this is not found in the prior edition.
- **Chapter 8.0 Outdoor Lighting Design Process**
 - **8.7.2.4 Color – Hue and Saturation:** Discusses considerations for lighting design, which is not present in the excerpts from the prior edition.

DO 064.

[Re-posting with clarification] References cards: 'Reference Marker' links should link to the place in the document where that superscripted endnote # is first referenced (usually the body of the document) instead of to the front cover or to the list of references at the back of the document.

Sample Search:

Provide a list of references in IES standards related to human vision.

Goal:

Ease of access

Example:

See below

LS-2-20 should link internally to **page 2 of LS-2-20** (see below).

As is often the case, power is more easily and accurately measured than is energy, and this is the case with radiant quantities. In this practical sense, luminous power (light, or luminous flux) is more fundamental or basic than luminous energy (the time-quantity of light). It should be noted that this definition is entirely different from the use of this term in physics, where it is synonymous with radiant energy of any wavelength.

illumination: This term is reserved to describe the general circumstance of light incident on a surface or

Radiometry and radiometric concepts deal with the measurement and conceptualization of radiant power as a physical entity; photometry and photometric concepts deal with visually evaluated radiant power. Photometric quantities always involve radiant power evaluated when the adaptation state is either photopic or scotopic. (Refer to *ANSI/IES LS-7-20, Lighting Science: Vision – Eye and Brain* for additional information.⁵) If there are parallel radiometric and photometric concepts, the same symbol is used, with the radiometric symbol being augmented with a subscript, e.

2

“Open in Library” should still link to G-1-22 page 79 where that reference was printed.

DO 075. [See also DO 047] If user is searching for a particular standard:

- In search field, use auto-suggest to return the name of a standard if they appear to be typing a match.
- If they search for a standard, return a “Document” card with the full title, description, thumbnail, and access to “Open in Library” or “Buy.”
- **Disable the AI Guide if user searches for a specific standard without any additional question/query.**

Sample Search: RP-3
Goal: Ease of access to documents
Example: See below

If user searches “ RP-3” then auto-suggest populates “ANSI/IES RP-3-26 Recommended Practice: Lighting Educational Facilities”.

Then it returns a result card with the document thumbnail, description and committee (see below).

CREATE NEW RESULT CARD FORMAT for searches by document title.

DO 070. Return these card formats only if the search term is a close or exact match for a document title or designation

DO 071. Accept variations on designation (all of the below are ‘acceptable’)

Sample Search: RP-3, RP-03, RP-3-20, RP-03-20, RP-3-20+E1, RP-03-20 E1...

DO 072. Present this card at top of search results

DO 073. Include thumbnail, description, and hyperlinked author (same data as from Table of Contents tool)

DO 074. Example:

The example result card is displayed within a light blue border. At the top left, a dark blue pill-shaped button contains the text "RP-3-20+E1". To its right, the text "Match: 55%" is shown. Further right is an orange pill-shaped button with a magnifying glass icon and the text "Open in Library". At the top right is a light blue pill-shaped button with the text "Document". Below these elements is a white rectangular box with a black border containing the text "Buy". The main content of the card begins with the title "ANSI/IES RP-3-26 Recommended Practice: Lighting Educational Facilities" in a bold, black font. Below the title is a small thumbnail image of a book cover with a blue and yellow design. To the right of the thumbnail is a paragraph of text: "Best practices to light classrooms and corridors, the many types of spaces in grade schools, high schools, and colleges including assembly halls, theaters, wet and dry labs, trade shops for woodworking or auto repair, and social spaces such as cafeterias and student and faculty lounges". Below this paragraph is a blue hyperlink: "IES Education, Library and Office Lighting Committee".

DO 076.

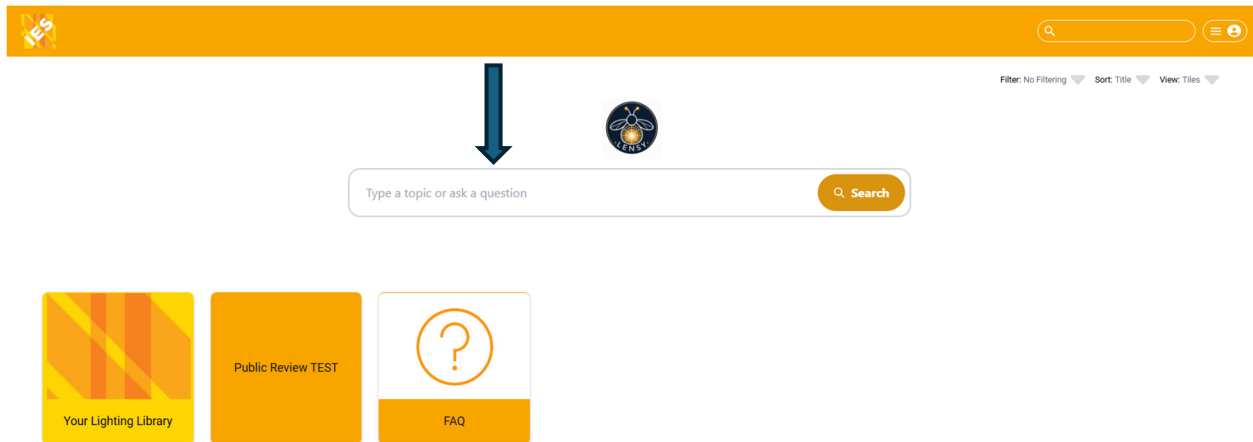
Vitrium: is it possible to generate an html embed code for Lensy that we can place in the Vitrium Portal Header ([Portal Settings](#) | [Vitrium Security](#))?

Clicking 'search' would open lensy.ies.org in a new tab and populate that query.

Goal: More intuitive way to access 'Lensy' search from within the Vitrium platform

Example: See below

Can we generate an html embed code for something similar to this?



DO 077.

If no results found, provide alternative paths. Offer guided recommendations for improving search experience (this would likely be a recommendation to broaden the search or clear filters, *or* ‘try rephrasing your question’ *or* ‘try searching by keyword or application’ or suggest spelling corrections, depending on the nature of the query.

Sample Search: See below

Goal: Provide feedback instead of “no results found”

Example: See below

Sample Search: what is the difference between luminance and illuminance?

- Filters: [only ‘Illum table’ selected]
- Results: ‘no results found’
- **Recommendation:** Try enabling the ‘Document Body’ filter and repeating your search.

DO 078.

Is it possible to continually ‘train’ the agent based on which cards people click on first to “open in library”, and perhaps which filters are engaged post-search for various search types?

Goal: Continually improve search experience over time and use

DO 079.

‘Compare Versions’ tool stayed ‘locked on’ after intentionally using it the first time, but intending to search without it the second time.

After a single search utilizing ‘Compare Versions’ deselect the tool (user needs to intentionally select it again each time they perform a ‘Compare Versions’)

DO 080.

Disable/Enable AI Guide tool: 'save' the state to the user's account. Every time the user returns to the tool, it should be in the same state it was last set.

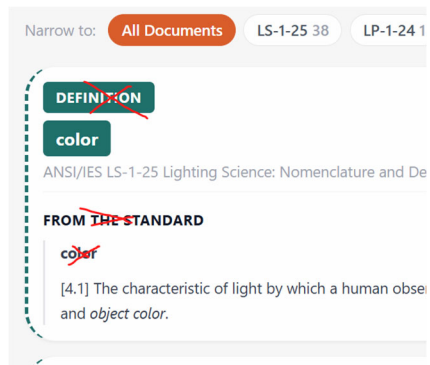
DO 081. Remove redundancy and simplify card structure.

Remove redundancy and simplify card structure.

Goal: A more ‘minimalist’ design without redundancy

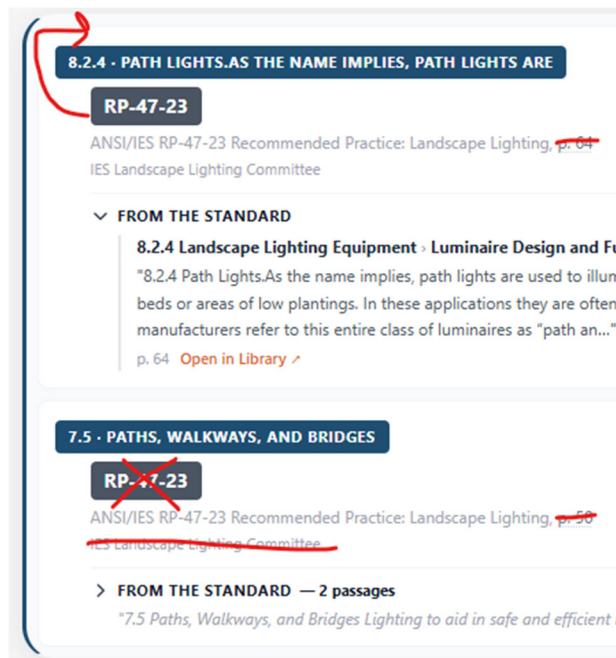
Example: See below

Definition Cards



- Delete “DEFINITION” label; it is self-evident.
- Delete “FROM THE STANDARD” and the repeated term; it is clear that the text (“[4.1] The characteristic...” is the definition of the term color.

Document Cards



- Move Document/title above chapter/section to clarify hierarchy
- For ‘joined cards’ don’t repeat Document/title, nor committee name
- The page number appears at end of each ‘From the Standard’ so we can delete the redundant instance beside the full title.

Reference Cards

The screenshot shows a digital reference card for the IES standard LM-83-23. At the top, the standard number 'LM-83-23' is highlighted in a dark blue box. Below it, the title 'ANSI/IES LM-83-23 Approved Method: IES Spatial Daylight Autonomy (SDA) and Annual Sunlight Exposure (ASE), ~~p. 36~~' is displayed, with the page number 'p. 36' crossed out in red. The publisher 'IES Daylight Metrics Committee' is listed below the title. A section titled 'REFERENCE ENTRY' with a downward arrow contains a text input field with the placeholder '"3. Clause (section), sub-clause or paragraph number; and page number: _____"' and a link below it that reads '\$R · p. 36 Open in Library ↗'. Below this is a 'REFERENCE MARKERS' section featuring a horizontal row of 14 purple pill-shaped buttons, each containing a standard number followed by a '3' (e.g., 'LM-20-20 3', 'LM-54-20 3', 'LM-80-21 3', 'LM-82-20 3', 'LM-83-23 3', 'LM-90-20 3', 'LM-92-22 3', 'LM-93-22 3', 'LP', 'RP-30-25 3', 'RP-42-26 3', 'TM-15-20 3', 'TM-21-21 3', 'TM-28-20 3'). At the bottom, a small line of text states: 'IES Standards that cite this reference — the link opens the page where the marker is first printed.'

- The page number appears at end of each 'From the Standard' so we can delete the redundant instance beside the full title.

DO 082. Provide a “View Table of Contents” option for each standard on the “List Standards” page. This could either be generated by indexing *or* could be pulled from the ‘preview’ files which we should be hosting for display on the webstore ([Previews](#)). Floating pdf viewer window?

Goal: Provide easy access to TOC for each document

Example: See below

Floating PDF viewer window?? Some TOCs are several pages long. Goal is to provide easy/free access to TOC, even if the user is on LensyLite (non-subscriber).

No cover

ANSI/IES RP-41-20
Recommended Practice: Lighting Theaters and Worship Spaces
62 pages Indexed
v Table of Contents

Read
Buy

Please refer to the IES Bookstore for possible Errata: www.ies.org.

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<u>DO 083.</u>	Document Comparison: Refine structure for greater clarity on how extensive the changes are and what topics/chapters those changes address. Length and specificity of response should <i>adapt</i> relative to the quantity and nature of the changes.
Sample Search:	What's new in the current version of RP-43-25?
Goal:	Greater depth of response (without overwhelming the user), but better clarity on whether changes are minimal or extensive.
Example:	See below

ADD section at top, which answers: how extensive are the substantive (*substantive = significant modification or expansion in the nature and scope of the content that could affect its application, results, or compliance with established standards*) differences between the 2 compared documents, and why might it matter?

1. **If extensive:** Provide **high-level** overview of 'new/updates/deletion' sections, organized by relevant chapters. Cap response at ~1500 words, with a typical response 800-1200 words.
2. **If moderate:** Provide a **concise summary of specific substantive changes** in 'new/updates/deletions' section, organized by relevant chapters. Cap responses at ~1200 words, with a typical response 500-1000 words.
3. **If minimal:** **List the substantive changes by topic** (not necessarily within the structure of 'new/updates/deletions'). Cap responses at ~500 words, with a typical response 100-300 words.

Hyperlink to specific chapters/pages in all instances.

Note: If the designation includes (R####) or (R##), then it is a 'reaffirmation' which means there are *no* differences between, for example, TM-31-20 and TM-31-20(R26). In that case, comparison would need to be between TM-31-20(R26) and TM-31-17 instead.

DO 084. AI Guide needs to present ‘disclaimer’ text for all instances of health/safety, public safety, fire code, and egress lighting.

Goal: Clarify that if your project involves emergency lighting, healthcare operating rooms, retail with energy limits, public safety lighting, or specialized spaces with jurisdiction-specific rules, you must coordinate with the AHJ to ensure compliance beyond IES standards

If user searches for any topics similar to:

1. Egress lighting
2. Life/safety or public safety lighting or active violent event policies
3. Healthcare operating rooms
4. Energy code requirements
5. Localized ordinances or codes or laws or lighting zone designations
6. Perimeter or security lighting
7. Other jurisdiction-specific policies

Then AI Guide should begin with a disclaimer similar to:

IES standards and guidance do not supersede applicable laws, codes, regulations, or project-specific requirements. For projects involving life safety, emergency systems, healthcare facilities, public safety, energy compliance, or other regulated applications, consult the applicable codes and coordinate with the Authority Having Jurisdiction (AHJ).

DO 085.

AI Guide: if user asks a question that is completely ‘out of scope,’ infers malevolent intent, doesn’t have any apparent answer in the indexed content, the AI Guide should not force a references to standards, and we should not see 50 results. Perhaps in this instance, the “Refine your search” prompt can be the first to state that no relevant results were found, and offer the user an opportunity to restate the question.

Sample Search: What color are zebras? *and* How can I use lighting fixtures to build a bomb?

Goal: Try that search...the results don’t inspire a lot of user confidence in the tool 😊

Example: See below

The screenshot shows a search interface with a search bar containing 'what color are zebras?' and a 'Search' button. Below the search bar is a yellow box containing an 'AI Guide' section. The 'AI Guide' section has a disclaimer: 'This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.' It then provides a detailed explanation of why the question is not directly addressed in the IES standards, referencing ANSI/IES LM-45-20(R2023) and ANSI/IES LS-1-25. It also includes a 'Further reading' section with links to ANSI/IES RP-30-25 and ANSI/IES LS-5-25. At the bottom of the yellow box, it says 'IES Lensy AI-Generated Summary — Not for reproduction'. Below the yellow box, there is a red circle around the text '50 Results' and a note 'for "what color are zebras?" · ordered by the AI Guide'.

what color are zebras? Search

AI Guide Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

The question of what color zebras are is not directly addressed in the IES standards, as these standards primarily focus on lighting and colorimetry in the context of human visual perception and illumination. However, we can explore how the IES standards approach color and colorimetry, which might provide some insight into the perception of zebra colors under different lighting conditions.

According to [ANSI/IES LM-45-20\(R2023\)](#), the standard references the International Commission on Illumination (CIE) publication CIE 015:2018, Colorimetry, which provides a foundation for understanding color and its measurement. This reference suggests that the IES standards rely on established colorimetry principles to inform their guidance on lighting and color. The definition of the Ostwald color system in ANSI/IES LS-1-25 provides further insight into how colors can be described and analyzed in terms of their content and purity.

The ANSI/IES LS-1-25 standard also includes Table T-5B, which provides color-matching functions and chromaticity coordinates for the CIE 1964 Supplementary Standard Colorimetric Observer. This table is relevant because it helps in understanding how colors are perceived by the human eye under different lighting conditions. While it does not directly answer the question about zebra colors, it contributes to the broader understanding of color perception, which is crucial in lighting design.

The IES standards referenced do not provide a direct answer to the question of what color zebras are, as this information is not within the scope of lighting standards. For accurate information on zebra colors, one would need to consult biological or zoological resources. However, for understanding how colors are perceived and measured in the context of lighting, the referenced standards, such as [ANSI/IES LM-45-20\(R2023\)](#) and ANSI/IES LS-1-25, offer valuable insights.

Further reading: [ANSI/IES RP-30-25](#) provides additional guidance on color difference formulas, which can be useful in understanding how colors are distinguished and measured in various lighting applications. This standard, along with others like [ANSI/IES LS-5-25](#), which references studies on the brightness of color, can enhance one's understanding of colorimetry and its application in lighting design. For specific questions not addressed by these standards, contacting Standards@ies.org can provide authoritative assistance.

IES Lensy AI-Generated Summary — Not for reproduction

50 Results

for "what color are zebras?" · ordered by the AI Guide

See also: *How can I use lighting fixtures to build a bomb?*

- Should not even attempt to provide IES references, despite the disclaimers
- Should not provide 50 card results

<u>DO 086.</u>	It doesn't seem to 'understand' tables and images as well as basic text. Are they also vectorized? Is it multimodal, and if not can it be?
Sample Search:	See below
Goal:	Allow users to find images, tables, and the content within them via effective search card results.
Example:	See below

Q: Where is the table that shows sound coefficients for various materials?

And “what is the sound absorption coefficient for linoleum?”

- Results direct user correctly to RP-1-24 Annex C, but do not cite Table C-1 (“Sound Absorption Coefficients for Various Materials”) on page 62 and do not appear to ‘know’ the ‘answer.’

Q: If I have a CSA/ISO Type 22 Monitor with positive polarity and 11ft spacing between uplights, what should be the maximum ceiling luminance gradient from uplights for critical surfaces?

- Search results do not include any references to the table on page 18 of RP-1-24, which contains a (rasterized) table specifically addressing this type of question.

Q: Where can I find images of what classifications for indoor fixtures look like visually? I need photometric web diagrams.

- Search results do not provide information directing users to where they can find those images. One such example is Figure 5-1 on page 26 of RP-1-24.

Q: Table of light loss factors

- A quality AI Guide response would be that there is not currently a consolidated table of light loss factors but that they are defined in full in RP-36 Section 3, and that this section does contain relevant tables...

DO 087.

Remove the visual “TM-24”, “Indoor” and “Outdoor” markers from the top right of all card types. ‘TM-24’ is a holdover from an old table format and is no longer needed on the result cards; ‘indoor’ and ‘outdoor’ is already communicated by “INTERIOR and “EXTERIOR” on the top left.

Sample Search: pickleball

Goal: Simplify

Example: See below

EXTERIOR - SPORTS

Pickleball

Class III > Area of play III

Cited by AI Guide

Outdoor

Area

TM-24

Match: 61% [Open in Library](#) [+ Save Search](#)

ANSI/IES RP-6-24 Recommended Practice: Lighting Sports and Recreational Areas, Table A-2, Row 175, p. 80
IES Sports and Recreational Area Lighting Committee

Horizontal (E_a) Cat. R

MAINTAINED	UNIFORMITY	BASIS
500 lx @ 0.91 m 50 fc @ 3 ft	2:01 Max:Min	Avg

> FROM THE STANDARD — Top 10 passages

"[Section C] Broomball See Ice H Miniature Golf Rec M 100 lx @ 0.91 m 3.0 ft Avg 0.25 3:01 Max:Min Horsehoes (general area) Rec K 50 lx @ 0.91 m 3.0 ft ..."

INTERIOR - SPORTS

Table Tennis

Cited by AI Guide

Indoor

Match: 60% [Open in Library](#) [+ Save Search](#)

IES Sports and Recreational Area Lighting Committee

Class III > Area of play III

Area

TM-24

ANSI/IES RP-6-24 Recommended Practice: Lighting Sports and Recreational Areas, Table A-1, Row 92, p. 74

Horizontal (E_a) Cat. R

MAINTAINED	UNIFORMITY	BASIS
500 lx @ 0.91 m 50 fc @ 3 ft	3:01 Max:Min	Avg

DO 088.

The first sentence in most ‘AI Guide’ results *tend* to be ‘fluff,’ restating the question without much substantive information. Since the user query remains visible, let’s jump straight to the substantive content.

Sample Search:

What’s an illuminance target for a LZ2 roadway

Goal:

Focus on substantive content and placing useful information at the top of each response.

Example:

See below

What's an illuminance target for a LZ2 roadway Search

AI Guide Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

For a LZ2 roadway, the illuminance target is a crucial aspect of lighting design. According to ANSI/IES RP-43-25, Lighting Exterior Applications, LZ2 is defined as a developed area with moderate anthropogenic light, where lighting is needed for increased mobility and convenience at night. This standard provides guidance on the application of lighting zones, including LZ2, and emphasizes the importance of considering the type of activity and the level of lighting required.

- 1st sentence does not add anything useful.

why does luminance matter? Search

AI Guide Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

Luminance plays a crucial role in lighting design as it affects the visibility of objects and scenes. According to ANSI/IES RP-1-24, luminance is a measurable quantity that results when light flux interacts with a surface, and it should not be confused with brightness, which is subjective and based on perception. The IES standards emphasize the importance of luminance in various applications, including office spaces and outdoor environments.

- 1st sentence does not add anything useful.

what is egress lighting? Search

AI Guide Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

Good Egress lighting refers to the illumination provided to ensure safe evacuation of a building or space during an emergency, such as a power outage. According to ANSI/IES RP-4-20, emergency egress lighting is required by most building codes to assist occupants in evacuating a building during a power outage. The lighting designer should check with the project team to determine the minimum maintained required lighting levels for emergency egress.

- This example is good.

DO 089. Introduce a “continue reading” element after approximately the first 50 words of the AI Guide.

what is the IES?

Search

what is the IES?

Search

AI Guide

Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

The Illuminating Engineering Society (IES) is a organization that develops and publishes standards for lighting. According to IES G-1-22, the IES administers a consensus standards development process approved by the American National Standards Institute, which brings together volunteers with varied viewpoints and interests to achieve consensus on lighting recommendations. This process is crucial in ensuring that the standards developed by the IES are comprehensive and effective.

-- Continue reading --

50 Results

for "what is the IES?" · ordered by the AI Guide

G-1-22

Cited by AI Guide

Document

Match: 64%

Open in Library

+ Save Search

IES G-1-22 Guide for Security Lighting for People, Property, and Critical Infrastructure, p. 4

IES Security Lighting Committee

FROM THE STANDARD

"DISCLAIMER IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development..."

p. 4 Open in Library

LM-63-19

Cited by AI Guide

Document

Match: 64%

Open in Library

+ Save Search

ANSI/IES LM-63-19(R25) Approved Method: IES Standard File Format for the Electronic Transfer of Photometric Data and Related Information, p. 2

IES Computer Committee

FROM THE STANDARD

"Copyright 2025 by the Illuminating Engineering Society. Approved by the IES Standards Committee November 27, 2024, as a Transaction of the Illuminating Engineering Society. Approved February 25, 2025, as an American National Standard. All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of ..."

p. 2 Open in Library

LM-73-18

Cited by AI Guide

Document

Match: 63%

Open in Library

+ Save Search

ANSI/IES LM-73-04(R18) Approved Method: For Photometric Testing of Entertainment Lighting Luminaires Using Incandescent Filament Lamps or High Intensity Discharge Lamps, p. 2

IES Testing Procedures Committee

FROM THE STANDARD

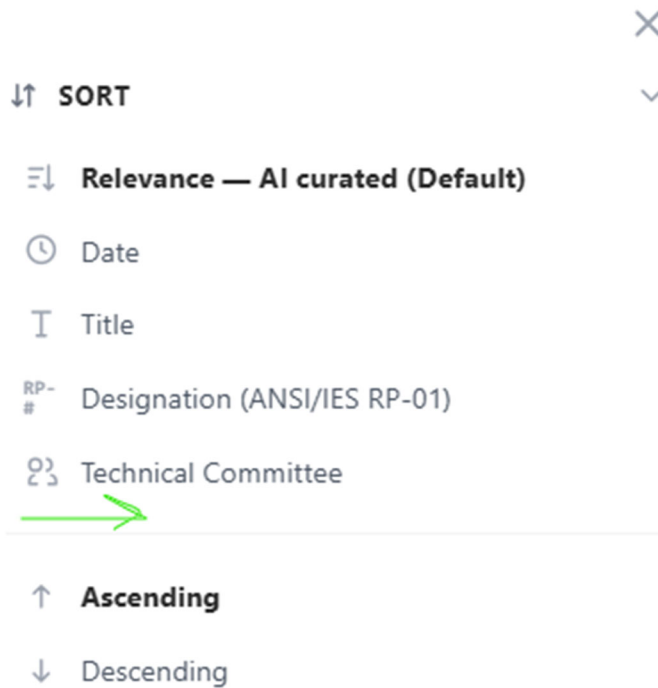
260820_Lensey Feedback.docx

26

DO 090. ADD 2 dynamic (post-search) filters, which only appear if search results include illuminance table cards containing these fields.

Sample Search: “walkway” or “tennis”

Goal: Ease of filtering



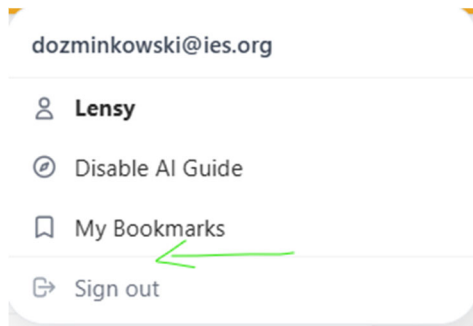
ADD FILTERS:

1. Lighting Zone Potential icon: 
2. Class of Play Potential icon: 

Note: ‘Class of Play’ is only relevant to RP-6 (sports) results

DO 091. Add feature in user profile drop-down to toggle primary units in illuminance table results from . This should *save* to their account until changed again.

Sample Search: Desk
Goal: User customization
Example: See below



Add toggle feature: **“Preferred units: SI or USCS”**

- Default should remain **SI (lux/meters)**.
- Toggling changes primary units to **USCS (footcandles/feet)**

DEFAULT (SI):

Horizontal (E _v) Cat. O			Vertical (E _v) Cat. M		
MAINTAINED	UNIFORMITY	BASIS	MAINTAINED	UNIFORMITY	BASIS
200 lx @ 0.91 m 20 fc @ 3 ft	3:1 Avg:Min	Avg	100 lx @ 0.91 m 10 fc @ 3 ft	3:1 Avg:Min	Avg

TOGGLED TO USCS:

Horizontal (E _v) Cat. O			Vertical (E _v) Cat. M		
MAINTAINED	UNIFORMITY	BASIS	MAINTAINED	UNIFORMITY	BASIS
20 fc @ 3 ft 200 lx @ 0.91 m	3:1 Avg:Min	Avg	10 fc @ 3 ft 100 lx @ 0.91 m	3:1 Avg:Min	Avg

DO 092.

The “See Notes” comment in RP-43 tables should be treated the same as a table footnote.

- Use red font
- Define it in drop-down “Application Task/Area Notes”

Sample Search: Walkway

Goal: Clarity

Example: See below

walkway Search

PEDESTRIAN SAFETY Cited by AI Guide Outdoor Area LZ4

Common Pedestrian Areas for Parks, Malls, Campuses, Commercial Spaces

Walking Surfaces (adjacent to architecture/exits/hardscape) › LZ4 › Lower limit (avg.)

Match: 66% Open in Library + Save Search

ANSI/IES RP-43-25 Recommended Practice: Lighting Exterior Applications, Table A-3, Row 99, p.78
IES Outdoor Nighttime Environment Committee

Horizontal (E _h) Cat. A		
MAINTAINED 10 fc	UNIFORMITY 5:1 Avg:Min	BASIS Avg

Environmental and Visual Considerations

LIGHTING ZONE LZ4	GLARE 5% Max	UPLIGHT 500 lm Max	SPECTRUM 2400K to 4000K
-----------------------------	---------------------------	---------------------------------	-----------------------------------

› FROM THE STANDARD — 6 passages

"8.6.1.1 Adjacent Walkways: Proper lighting of pedestrian the area of high luminance into darker surroundings. walkways is essential to their safe use b..."

- Change font color of “See Notes” to red

› Application Task/Area Notes — footnotes begin collapsed. Click to display.

- Add to Application Task/Area Notes drop-down

The text for these “See Notes” is found here:

Table A-3. Recommended Illuminance Criteria for People in Outdoor Environments

Application Task/Area Notes	
1	Data fields with no value indicate there is no expectation of needing a recommendation for that application or task.
2	For average illuminance, a range is recommended instead of a single value. The values shown are the lower and upper ends of this average-illuminance range. The choice of illuminance level within a range will vary according to the conditions of each site and task. Refer to Table A-1 for guidance in selecting the illuminance target value.
3	If an area is environmentally sensitive, retroreflective materials should be considered before adding area lighting.
4	Outdoor lighting carries a responsibility to balance human and environmental needs. When human vision is challenged the most, the quality of the light provided is often more important than the quantity of light. Refer to Section 5.0 for additional recommendations regarding responsible outdoor lighting.
5	Glare (max): Using the luminaire classification system (LCS) found in ANSI/IES TM-15, a maximum percentage of light emission just below horizontal (FVH) is recommended for luminaires emitting greater than 2,000 lumens. To be useful and beneficial, light will need to be targeted for the task and avoid creating potential glare or wasted emissions. See Section A.2.2 for additional guidance.
6	Uplight (max): Light emission limits are recommended for uplight, as measured by the total number of lumens emitted within the Uplight Low-Angle (UL) zone of the TM-15 Luminaire Classification System. See Section A.2.2 for additional guidance.
7	Controls: Directing light to hit its target is not sufficient to minimize ecological impacts on nocturnal species and habitat. Every effort should be made to reduce light when it is not needed. Extinguish or dim the light when it is not needed during times of vacancy, curfew, or seasonal conditions. See Section A.2.2 for additional guidance.
8	Controls: Illuminance reductions during vacancy, curfew, and nighttime hours may not be recommended when there are pedestrian safety and/or security reasons. See Section A.2.2 for additional guidance.
9	Controls: Using controls to dim or turn off lighting during vacancy, curfew, and seasonal conditions is represented by a percentage of full light output. For example, "20%" refers to 20% of full output. See Section A.2.2 for additional guidance.
10	Spectrum: There is not yet widespread agreement on a more relevant metric than CCT for outdoor spectral considerations. For this reason, CCT is used here. CCT values available for consideration are: CCT ≤ 1800 K; CCT ≤ 2400 K; CCT ≤ 3000 K, and CCT ≤ 4000 K. See Section A.2.2 for additional guidance.

DO 093.

In 'Illuminance Table' cards (only), change “FROM THE STANDARD” to “FROM THE STANDARD (browse relevant design considerations):”

Goal: Encourage users to explore more than the quantitative recommendations in the tables

Example: See below

tennis

ANSI/IES RP-6-24 Recommended Practice: Lighting Sports and Recreational Areas, Table A-1, Row 95, p. 75

Horizontal (E _h) Cat. U		
MAINTAINED 1500 lx @ 0.91 m 150 fc @ 3 ft	UNIFORMITY 1.5:1 Max:Min	BASIS Avg

▼ **FROM THE STANDARD** — Top 10 passages

6.13 Reason and substantiation › Handball, Racquetball, and Squash

Change to:

FROM THE STANDARD (browse relevant design considerations) – Top 10 passages

DO 092. Not sure ‘why,’ but this particular search produced results that seemed poorly prioritized.

Sample Search: Recommended illuminance level for sidewalk in a zone 1 in a rural environment?

Goal: Well-prioritized search results

Example: See below

Actual AI Guide order	Actual Card results order	IDEAL order for both
LP-2	G-1	RP-43 FIRST
LP-1	LP-1	RP-8 SECOND
G-1	RP-43	Others to follow
RP-8	RP-8	
RP-43	LP-2	

This question is solidly within the scope of RP-43 (Lighting Design for Outdoor Pedestrian Applications), and Lighting Zones are heavily referenced in RP-43, so RP-43 should have been the most prominent reference.

It is possible that “sidewalk” was not associated with critical adjacent terms in RP-43, such as:

- Wayfinding
- Paths
- Pedestrian Malls
- Pedestrian Tunnel
- Walking Surfaces
- Building Entrances

Pedestrian-oriented applications will be a common search term; let’s do what we can to make sure context is ‘understood’ here.

DO 093.

Add several critical pages from the IES website to indexing sources, for purposes of AI Guide context. *Lensy should never 'contradict' the language in the specific sources listed below.* Provide links to these pages *if* referenced.

Sample Search:

See examples below

Goal:

Include context from position statements, IES mission, etc to the way in which AI Guide statements are drafted when relevant.

ADD the following to indexing for reference:

1. [About the IES](#) (only these 2 items):
 - a. Mission Statement
 - b. Vision Statement
2. [Advocacy](#) (Position Statements only, which may change over time):

Position Statements

IES Position Statements are approved by the [Society's Board of Directors](#) and reflect the Society's official position on lighting-related or the general public interest. They are often interpretations of content or use for IES Standards, however, they can also represent issues generated outside of the Society. These position statements are developed by technical experts within our [Standards Committee](#) and are free to use and disseminate.

> PS-06-24: IES Position on Legislation and Ordinances for Outdoor Light at Night

[Read PS-06-24](#)

> PS-13-21: IES Position Statement on NEMA LS 20001-2021, White Paper on Unified Glare Rating (UGR)

[Read PS-13-21](#)

> PS-12-19: IES Position On UL RP 24480 Regarding Light and Circadian Entrainment

[Read PS-12-19](#)

> PS-11-18: IES Position on TM-30-18, IES Method for Evaluating Light Source Color Rendition

[Read PS-11-18](#)

> PS-10-18: IES Position on LED Product Lifetime Prediction

[Read PS-10-18](#)

> PS-09-17: IES Board Position on AMA CSAPH Report 2-A-16, Human and Environmental Effects of Light Emitting Diode (LED) Community Lighting

[Read PS-09-17](#)

a.

Sample searches that would need to return relevant content in the AI Guide:

- What is the IES?
- Is the MLO still valid?
- What should legislation include regarding outdoor nighttime lighting?

DO 094. Let's finesse the quantity of search card results populated, based on quality and relevance – particularly when AI Guide is enabled.

Rules (feel free to modify based on best practice):

1. It's ok for a search to return less than 50 cards when there is no (or very limited) relevant content.
2. It's ok for a search to present far more than 50 cards when there are a large quantity of results which *may* be relevant.
3. If >10 cards presented and cards are sorted by relevance: When AI Guide is enabled, provide a "Show more results" style element when AI Guide is enabled, after cards with a high relevance confidence are displayed.
 - a. 'AI Guide' should determine relevance of search cards by considering "is it likely that the user will find this card useful?" Incorporate the "Show more results" element when cards with that level of confidence have been depleted for the rest of the list.

DO 095.

Any time a user asks a question about a deprecated standard, the AI Guide should begin by stating that the standard is deprecated.

Sample Search: *What does RP-20 cover?*

Goal: Transparency

Example: See below

Search

 **AI Guide**

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

The query about RP-20 does not yield a direct answer from the search results provided. However, we can explore the related standards and their coverage to understand how the Illuminating Engineering Society (IES) approaches similar topics. The search results include various IES standards such as **ANSI/IES RP-2-20+E1**, **ANSI/IES RP-3-20+E1**, **ANSI/IES RP-4-20+E1**, and **ANSI/IES RP-43-25**, which cover lighting practices for retail spaces, educational facilities, library spaces, and exterior applications, respectively.

Beta

The best response would be a brief statement:

“ANSI/IES RP-20-14 is deprecated. Refer directly to the deprecated standard for historical purposes.”

Or, if the user asks a question about a deprecated standard which has a current version:

“ANSI/IES RP-43-22 is deprecated and has been replaced by the current **ANSI/IES RP-43-25 Recommended Practice: Lighting Exterior Applications.”**

DO 096.

Any time a user asks a question about a particular lighting application, if there is an RP (Recommended Practice) with that application within its scope then it should be cited *first*. G-1 should never be cited first unless it is the only standard that addresses an application, or if the user asks about ‘safety.’

ALSO: Make sure it is understood that RP-8-22 is a deprecated version of RP-8-25+E1 and RP-8-25+E2.

Sample Search:

What documents provide lighting levels for parking garages

Goal:

Clarity

Example:

See below

Search

 **AI Guide**

Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

When it comes to lighting levels for parking garages, the Illuminating Engineering Society (IES) provides guidance through several standards. According to ~~IES G-1-22, Guide for Security Lighting for People, Property, and Critical Infrastructure~~, recommendations for lighting parking lots and garages can be found in ANSI/IES RP-8-22, Recommended Practice: Lighting Roadway and Parking Facilities. However, since ANSI/IES RP-8-22 is not available in the search results, we can refer to ANSI/IES RP-8-25+E1 and ANSI/IES RP-8-25+E2, which provide similar guidance.

- Cite RP-8 first
- It should not cite RP-8-22, as that is deprecated and replaced by RP-8-25+E2.
 - o *Note: _+E# indicates an “errata” and the highest +E# is therefore the current version of a standard.*

DO 095. AI Guide: make the “Continue Reading” element slightly more prominent, as it blends in to the appearance of all other hyperlinked standards.

How do IES standards address lighting for older people? **Search**

How do IES standards address lighting for older people? **Search**

AI Guide Beta

This AI-generated response is for informational purposes only and may contain errors. Always refer to the full IES Standards for authoritative guidance. This response cannot be saved to your search collections.

The Illuminating Engineering Society (IES) standards address lighting for older people through the **ANSI/IES RP-28-25** Recommended Practice: Lighting and the Visual Environment for Older Adults and the Visually Impaired. This standard provides guidance on designing lighting environments that cater to the needs of older adults, taking into account their visual abilities and requirements. According to **ANSI/IES RP-28-25, Section 1.0** Introduction and Scope, older adults represent the fastest growing segment of the population, and their visual needs must be considered in lighting design.

Continue Reading

IES Lensy AI-Generated Summary — Not for reproduction

Slightly larger font? Present it in a ‘pill’ button?

DO 096.

AI Guide: Keep “Further Reading” bolded, but don’t need to bold the rest of the paragraph. This should be *slightly* less prominent than all the highly-relevant entries above.

Sample Search:

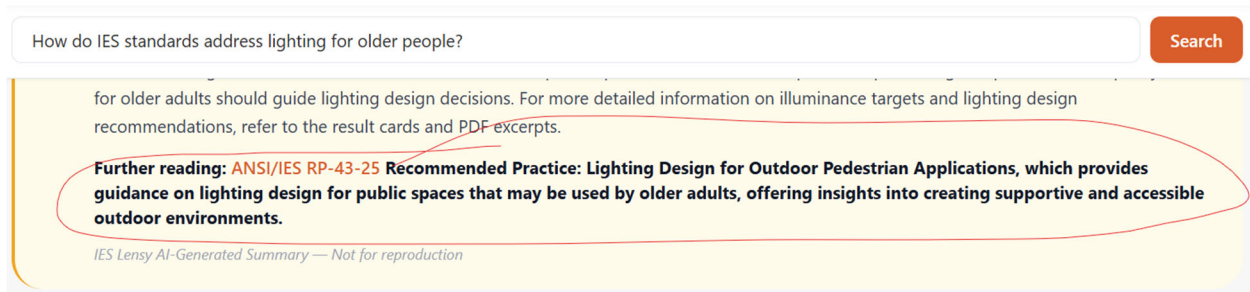
How do IES standards address lighting for older people?

Goal:

Hierarchy; prioritize most relevant information only

Example:

See below



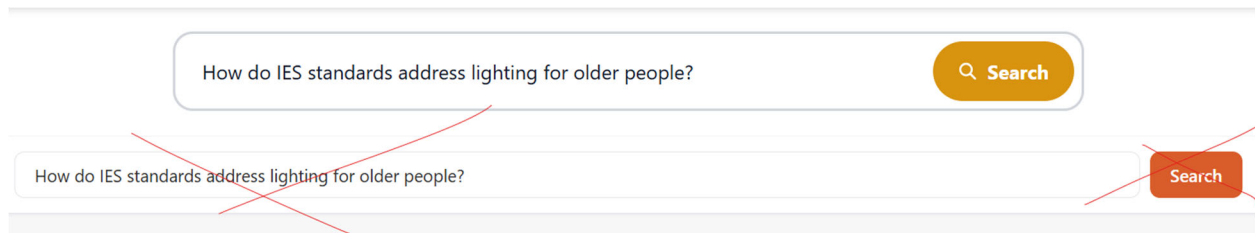
- Keep “Further Reading” bold
- Rest of text: not bold

DO 097. Only display 1 search bar at a time (users found it confusing to see 2, with the same text in each)

Goal: Clarity, simplicity

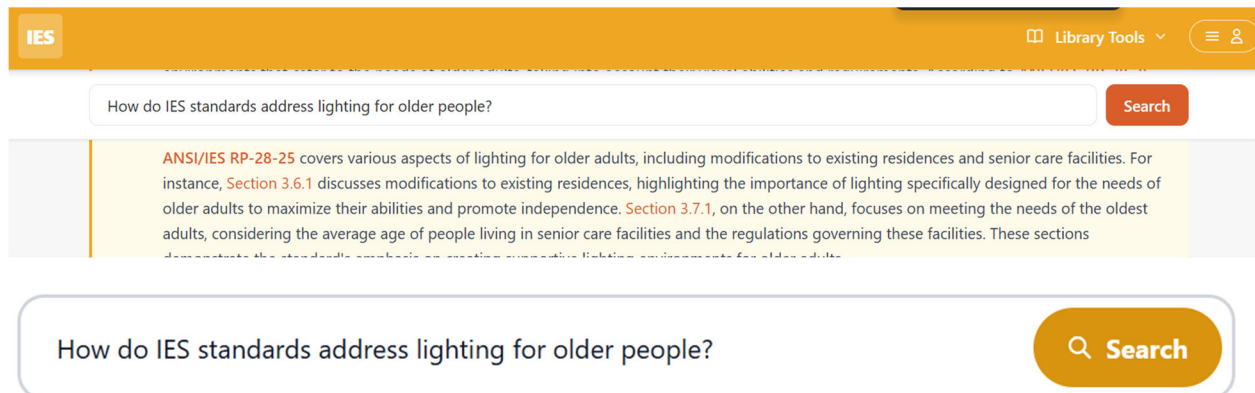
Example: See below

At top of page:



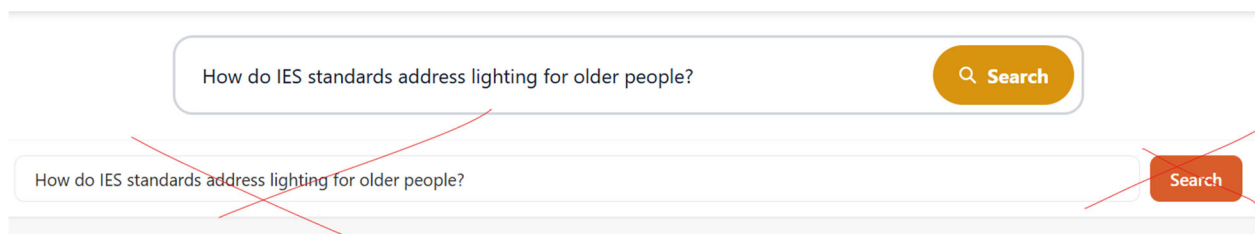
- Don't show the 'floating' bar until the original one is no longer visible

When you scroll down:



- Good to show the floating one, but keep the same style (color, shape and icon) of the original search bar

If you scroll back to the top of the page:



- (same as above) still just see 1 search bar